



ST BENEDICTS COLLEGE
NOVEMBER EXAMINATIONS

ADVANCED PROGRAMME MATHEMATICS GR 11

Examiner: Mrs D Park-Dobell

Time: 2 hrs

Moderator: Mrs M Povall,

Total: 200

QUESTION NO	DESCRIPTION	MAXIMUM MARK	MARK OBTAINED
1	ALGEBRA	32	
2	PARTIAL FRACTIONS/COMPLEX NOS	29	
3	SPLIT GRAPHS	22	
4	COMPOSITE FUNCTIONS	21	
5	RADIAN MEASURE	8	
6	VOLUME/RADIAN MEASURE	13	
7	TRIGONOMETRY	22	
8	DIFFERENTIATION/LIMITS	17	
9	FIRST PRINCIPLES/ AVE GRADIENT	16	
10	CALCULUS.	20	
	TOTAL	200	

INSTRUCTIONS TO CANDIDATES



- ENSURE YOUR CALCULATORS ARE IN RADIAN MEASURE
- ROUND OFF ALL ANSWERS TO 3 DECIMAL PLACES.

QUESTION 1

Solve for x if:

a) $\frac{x+4}{x^2+6x+8} - \frac{x+3}{x+2} = \frac{1}{x+4}$ (8)

b) $\frac{-32}{x+2} = \frac{4}{\sqrt{x+2}} - 1$ (8)

c) $\frac{36-x^2}{x} \leq 0$ (8)

d) Determine algebraically the solution of $|2x + 1| \leq 4$.

Illustrate, using a graph, where this solution can be read off. (8)

[32]

QUESTION 2

a) If it is given that $f(x) = \frac{x^2+4x+3}{x+2}$, show that $f(x)$ can be written as $x + 2 - \frac{1}{x+2}$ (5)

b) Split $\frac{5}{x^2-9}$ into its partial fractions (8)

c) Given $p(x) = x^3 + ax^2 + bx - 6$ with a zero at $x = 1 + i$. Determine the values of a and b (8)

d) Determine the values a and b , where $a, b \in \mathbb{R}$, and satisfy the equation

$$\frac{(a+3i)}{(2-5i)} \times bi = -11 - 13i \quad (8)$$

[29]

QUESTION 3

$$g(x) = \begin{cases} \frac{1}{x} & \text{if } x < 0 \\ x^2 - x & \text{if } x \in [0; 2] \\ 2 & \text{if } x > 2 \end{cases}$$

a) Sketch the graph of $g(x)$ showing all turning points and intercepts with axes (8)

b) Discuss, with motivation, the continuity of

$g(x)$ at:

i) $x=0$ (4)

ii) $x=2$ (4)

c) Give all values of x for which:

- i) $g(x) > 0$ (2)
 - ii) The function is increasing (2)
 - iii) The function is decreasing (2)
- [22]

QUESTION 4

a) If $f(x) = \frac{1}{3}x + 3$ and $x \in [-3:6]$, determine f^{-1} , state the new domain and range and draw a sketch graph of both graphs on the same set of axes. (9)

b) Fill in the missing functions in the table below. (6)

$g(x)$	$f(x)$	$(f \circ g)(x)$
$x + 2$	$3x$	
	$\sqrt{x - 5}$	$\sqrt{x^2 - 5}$
$\frac{1}{x}$		x

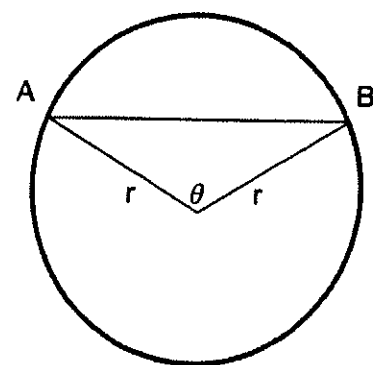
c) If $F(x) = \sin^4 4x = f \circ g \circ h$, determine f, g and h (6)

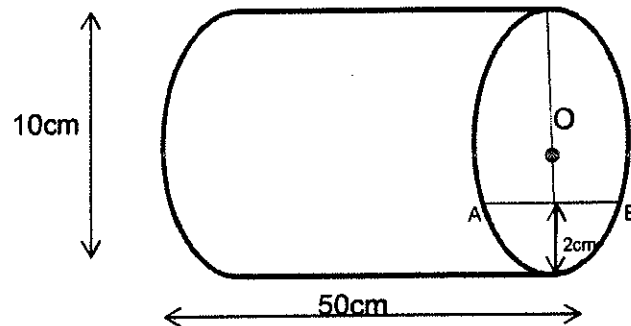
[21]

QUESTION 5

The chord AB of a circle of radius r subtends an angle of θ radians at the center O of the circle.

- a) Find an expression for the area of the shaded region in terms of r and θ (3)
 - b) Evaluate this expression when $r = 2$ and $\theta = \frac{2}{3}\pi$ (3)
 - c) Determine the minor arc length AB (2)
- [8]



QUESTION 6

A cylindrical pipe with a diameter of 10 cm and length of 50cm is lying on its side and there is water inside to a depth of 2cm.

- a) Show that the width of the surface of the water (length AB) on the diagram is 8cm (4)
- b) Hence find the volume of water in the pipe. (9)

[13]

QUESTION 7

- a) If $\cos 20^\circ = t$, find in terms of t : (12)
 - i) $\sec 200^\circ$
 - ii) $\operatorname{cosec} 160^\circ$
 - iii) $\tan 200^\circ$

- b) Prove that $\left(\frac{1}{\cos\beta} - \tan\beta\right) = \sqrt{\frac{1-\sin\beta}{1+\sin\beta}}$ (10)

[22]

QUESTION 8

- a) Determine $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ (4)

Determine $\frac{dy}{dx}$ in each of the following:

- b) $y = 3x^2 - 5x + 2$ (2)
- c) $y = \frac{x^3 - x^2 + x}{\sqrt{x}}$ (6)
- d) $y = \frac{x^3 - 3x + 2}{x - 1}$ (5)

[17]

QUESTION 9

$$g(x) = 3x^2 - 6x$$

- a) Determine $g'(x)$ by first principles (8)
 - b) Find the average gradient of the curve between $x = 1$ and $x = 3$ (8)
- [16]

QUESTION 10

$$\text{Given } f(x) = x^3 - 3x^2 + 4$$

- a) Determine $f'(x)$ (3)
 - b) Determine the equation of the tangent to $f(x)$ at $x = 1$ (8)
 - c) If $g(x)$ is the anti-derivative of $f(x)$. Determine a possible equation for $g(x)$ (7)
 - d) Explain why the word 'possible' is used in question c) above. (2)
- [20]

